# Ocular and Oral Complications

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## 48.1 Ocular Complications

### 48.1.1 Introduction

Ocular complications in HCT patients include both ocular GVHD- and non-GVHD-related late effects which, if left untreated, can lead to ocular discomfort and blindness. Non-GVHD-associated complications include cataracts, glaucoma, infections particularly (viral retinitis, fungal endophthalmitis), and posterior segment abnormalities. Ocular GVHD (oGVHD) is a rapidly progressing highly inflammatory condition that mainly affects the ocular surface and adnexae, but also the posterior segment in rare cases.

The World Health Organization considers blindness as one of the top causes of disability-adjusted life years, and it significantly affects the QoL. Thus, prevention and prompt management of ocular complications should be a priority of a transplant program given the morbidity associated with it.

### 48.1.2 Ocular GVHD

Ocular GVHD occurs as acute, chronic, and overlapping forms with chronic oGVHD representing the most common form with an onset between 5 and 24 months after allo-HCT (Kitko et al. 2021; Wolff et al. 2021).

The occurrence of oGVHD is variable in different series but is approximately 40–60% of patients receiving HCT (Nassar et al. 2013). The risk factors for oGVHD include donor-recipient HLA and gender disparity (female donor to male recipient), an older donor age, preexisting diabetes, and environmental stress (Gehlsen et al. 2022a).

The 2020 NIH consensus development project on cGVHD describes distinct differences of oGVHD from the dry-eye disease such as rapid manifestation, activation of donor hematopoietic/mesenchymal stem cells, early fibrosis, and potential intraocular involvement, although clinical phenotypes may be initially similar. The NIH cGVHD consortium published diagnostic best practice components that included eye care exams within a month prior to hematopoietic cell transplantation or within 3 months afterward and at regular intervals (e.g., every 3 months) through an ophthalmologist. These exams should include

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surveillance for infection, cataract formation, and increased intraocular pressure for prevention of ocular complications in GVHD.

The principles of oGVHD *management* include lubrication, drainage control, evaporation control, minimization of ocular surface inflammation, and improving epithelial wound healing. Based on the 2014 NIH chronic GVHD consensus panel, current recommendations for stepwise treatment of oGVHD include preservative-free artificial tears or ointments as a basic therapy, together with early application of topical cyclosporine and/or steroid eye drops for mild/moderate oGVHD. For severe forms, dosages are increased and serum eye drops and contact lenses are added (Carpenter et al. 2015). Among surgical procedures for moderate/severe oGVHD, amniotic membrane transplantation, punctal occlusion, and partial tarsorrhaphy have been recommended There is also some evidence on other modalities of treatment which include occlusive eye wear, lid care/warm compresses, and humidified environment. Corneal melting, superinfection, and perforation should be avoided at all costs as corneal transplantations frequently fail due to graft rejection and disturbed wound healing.

In addition, special attention should be given to other risk factors for ocular complications.

### 48.1.3 Posterior Segment Complications

Retinal and vitreous hemorrhages are not uncommon in HCT patients and may happen with or without the presence of oGVHD (Yoo et al. 2017). This is complicated by the presence of thrombocytopenia early in transplant but also later in the course since both drugs and chronic GVHD can be associated with thrombocytopenia. Prompt referral to an ophthalmologist is the key to preventing blindness; therefore, the practicing transplant clinician should have a high suspicion of retinopathy, retinal tears, or vitreous hemorrhages when a patient complains of "floaters" or just "decreased vision," which happens suddenly.

### 48.1.4 Ocular Infections

CMV infection is one the most widely studied ocular infections and can rapidly lead to retinitis, and since quite often IV or intraocular drug applications are required, prompt referral to an ophthalmologist is mandatory. Apart from CMV, adenovirus is also a common virus and can lead to viremia if untreated. Moreover, unlike immunocompetent individuals, varicella zoster infection within hours or a couple of days can lead to dissemination as well as postherpetic neuralgia, cranial nerve palsies, zoster paresis, meningoencephalitis, cerebellitis, myelopathy, and irreversible blindness.

Fungal infections in severely IS HCT patients (particularly those on multiple IS for GVHD) can quickly lead to mortality; thus, prompt referral for IV antifungals is indicated. Aspergillosis, mucormycosis, and candida have been reported in GVHD patients affecting the ocular tissues.

### 48.1.5 Glaucoma

Since the most common subtype of glaucoma (primary open-angle glaucoma) presents with gradual symptoms, its diagnosis is frequently missed in early phases. However, many risk factors in HCT can predispose to glaucoma and can lead to blindness which includes diabetes (allo-HCT patients have a four times higher risk of diabetes), retinopathy, and steroid use (for GVHD). Since the diagnosis of glaucoma is based on tonometry, gonioscopy, perimetry, and ophthalmoscopy, regular screening by the ophthalmologist is indicated.

### 48.1.6 Cataract

Cataract is the most common cause of blindness in the developed world. Risk factors in the HCT patients include steroid use, total body irradiation, and diabetes. Since intraocular lens implantation (particularly via phacoemulsification) has become a widely performed procedure worldwide for the treatment of cataracts, early recognition and prompt treatment can help in the preservation of vision. Although generally associated with a low rate of complications in oGVHD, complication rate can be elevated (Gehlsen et al. 2022b). Therefore, special attention should be applied perioperatively and prevention of cataracts by controlling risk factors should be a management strategy in HCT survivors if possible (Tables 48.1 and 48.2).

**Key Points**

- Ocular complications of HCT are not restricted to GVHD, since both non-GVHD allo-HCT and auto-HCT recipients can suffer from cataracts, viral/fungal infections, glaucoma, and retinopathies.
- Early detection and aggressive therapy by ophthalmologists familiar with ocular GVHD are paramount due to ensuing blindness.
- Transplant centers should consider a close collaboration with ophthalmology teams both for treatment and preventative strategies. This could ideally be achieved in a multidisciplinary team in a long-term follow-up or a survivorship clinic.

**Table 48.1** Ocular complications of HCT

| | Risk factors/manifestations | Management | Survivorship issues |
|---|---|---|---|
| GVHD-associated | – Acute and chronic ocular GVHD with dry-eye phenotype<br>– Retinopathies<br>– Cataracts<br>– Infections | – Topical: artificial tears, steroids, CSA, steroids, scleral lenses, serum eye drops, surgical procedures<br>– Systemic: ECP, rituximab, ibrutinib | Presence of ophthalmologist preferentially pre-allo-HCT and in the long-term follow-up or survivorship clinic |
| Non-GVHD-associated | – Glaucoma<br>– Cataract<br>– Infection<br>– Retinal/vitreous hemorrhage | – Removal of risk factors<br>– Surgery<br>– Antimicrobials | |

**Table 48.2** Oral complications of HCT

| | Risk factors/manifestations | Management | Survivorship issues |
|---|---|---|---|
| GVHD-associated | Risk factors; poor oral hygiene<br>Manifestations:<br>– Lichen planus-like patterning<br>– Hyperkeratosis<br>– Dry mouth<br>– Oral ulcers<br>– Oral sensitivity<br>– Mucoceles<br>– Mucosal erythema or atrophy<br>Long-term risks: oral squamous cell carcinoma, salivary gland dysfunction | – Topical: steroids of progressive potency, tacrolimus for lip lesions<br>– Systemic: steroids, ruxolitinib, belumosidil, extracorporeal photopheresis,<br>– Physical therapy for reduced oral opening<br>– Dry mouth: palliative aids, topical fluoride<br>– Regular dental care | – Regular dental care<br>– Regular oral cancer screening<br>– Participation of a dentist with HCT expertise in the long-term follow-up or survivorship clinic |
| Non-GVHD-associated | – Dysgeusia, temporomandibular joint disorders, mucositis, dental caries | | |

## 48.2 Oral Complications

### 48.2.1 Introduction

Oral complications in HCT patients include both oral GVHD and non-GVHD-related late effects which can adversely affect patient nutrition, quality of life (QoL), and health.

### 48.2.2 Oral GVHD

Three domains of the craniofacial complex are impacted by oral cGVHD: oral mucosa, salivary glands, and oral opening. Symptoms often begin with new-onset dry mouth and sensitivity to spice and mint flavors. At the onset, oral cGVHD may appear mucositis-like, but mature lesions can progress to tissue fibrosis and restrict oral opening and oral function and lead to permanent salivary gland dysfunction.

A diagnosis of oral cGVHD can be made clinically or with the aid of an oral mucosal biopsy. Diagnostic clinical features of oral cGVHD include the presence of lichen planus-like features which resemble white lacelike hyperkeratosis and may occur on any oral mucosal surface (Jagasia et al. 2015). Distinctive features include erythema, ulceration, oral dryness, mucoceles, mucosal atrophy, and pseudomembranes. In the case of distinctive features only, other potential causes of these lesions should be diagnostically excluded per the 2014 National Institutes of Health chronic GVHD consensus criteria (Jagasia et al. 2015). Other manifestations of oral cGVHD include low saliva production, gingivitis, oral sensitivity, oral pain, and mucosal fragility. These can lead to dental caries and chronic ulcers. Sclerosis of the perioral skin, fibrosis within the cheek tissues, and GVHD manifestations in the temporomandibular joints can reduce oral opening which impairs oral hygiene and nutrition and may impede oral access for necessary medical and dental procedures. Finally, avascular necrosis (AVN) of the temporomandibular joint can occur as a complication of steroid therapy for GVHD (Treister et al. 2012).

The 2020 National Institutes of Health chronic GVHD consensus criteria recommend oral exam prior to transplant and at day +100 that includes assessment of linea alba, lichen planus-like changes, and mucosa abnormalities (Kitko et al. 2021). Regular oral exams starting at day +100 posttransplant should include evaluation for any lichen-planus like changes, ulcers, erythema, and restriction of mouth opening. This exam should include an inquiry about patient-reported pain, difficulty swallowing, or oral dryness. Identification of new lesions suggestive of oral GVHD as detailed above should be followed with a referral to a dentist with experience in dealing with long-term HCT complications, often an oral medicine specialist.

The *management of oral GVHD* may require both topical treatments and systemic therapy. Topical treatments may include topical steroids of increasing potency as needed and calcineurin inhibitors on the vermillion lip where steroids are contraindicated. Emerging topical therapies include photobiomodulation and autologous blood products (Baumerin et al. 2022). Adjunctive medications for pain and symptom management are recommended to improve oral intake and QoL. Physical therapy may help to restore and maintain oral function. Among nonsteroidal systemic treatments for oral GVHD, good oral GVHD response rates have been reported for ruxolitinib, belumosudil, and extracorporeal photopheresis (Kaurinovic et al. 2022; Jagasia et al. 2021; Malik et al. 2014).

### 48.2.3 Subsequent Oral Cancers

Long-term survivors of HCT are at risk of subsequent cancer development. Oral squamous cell carcinoma is the most frequent subsequent cancer of the oral cavity (Inamoto et al. 2015). Risk factors include transplant for nonmalignant disease and GVHD with the highest increased risk in patients with a history moderate–severe oral GVHD and in those with >15-month duration of oral GVHD (Santarone et al. 2021; Schaar et al. 2021). Accordingly, aggressive and early treatment of oral GVHD is essential to reduce risk. Oral exam including an oral cancer screening is recommended at least every 6 months while HCT patients remain on immunosuppression

(Carpenter et al. 2015). Treatment of these lesions often includes surgical resection and adjuvant therapy in cases with high-risk pathologic features (Hanna et al. 2018). Lifestyle factors that contribute to oral cancer risk include radiation exposure, tobacco use, alcohol intake, and betel nut chewing and should be avoided in HCT survivors.

### 48.2.4 Non-GVHD-Associated Oral Complications

HCT patients with reduced immunity have increased susceptibility to oral infections. New lesions in the oral cavity should be screened for viruses including herpes simplex virus, fungal involvement including *Candida albicans* overgrowth, and drug reactions. Patients on systemic mTOR inhibitors may develop mTOR inhibitor stomatitis when blood levels of drug are elevated.

Salivary gland dysfunction may occur independent of GVHD following HCT and contributes to the rapid progression of dental decay and reduced QoL. Patients with myeloablative conditioning regimens frequently present with salivary gland dysfunction in the posttransplant period. Loss of saliva production may result from a history of total body irradiation or targeted irradiation due to progressive irreversible fibrosis of the acinar units and ducts within the salivary glands. Similarly, temporary reduction in saliva production may result from the use of specific medication classes including antidepressants, immune checkpoint inhibitors, antihistamines, opioids, and others. Dry mouth, or xerostomia, may be reported by patients if the quality or mucous content of saliva decreases even if the rate of its production remains steady.

Dysguesia following HCT is not well understood. Taste disturbances in both the flavor and texture experience are associated with specific chemotherapy regimens, mucosal damage, and alteration in the oral microbiome as well as with GVHD onset and generally result in lower caloric intake and reduced QoL in patients (Scordo et al. 2022). Dysguesia in most patients resolves with time.

### 48.2.5 Dental Requirements of the HCT Patient

Dentists with expertise in oral medicine should be part of the multidisciplinary team managing late effects in both autologous and allogeneic HCT survivors as endorsed by the National Institutes of Health's Late Effects Initiative (Hashmi et al. 2017). Once the primary indication for HCT is cured, it is essential that surveillance and preventative strategies be undertaken to alleviate the burden of comorbidities in these survivors. Oral hygiene and health are important for these patients as mucosal inflammation from gingivitis and periodontitis increases susceptibility to infection, with a less clear relationship with GVHD risk (Williams et al. 2021). Regular oral cancer screening is required, and intensive management may be needed for GVHD and non-GVHD oral complications after HCT. Thus, a long-term follow-up clinic should optimally have dentistry services available ad hoc if not on a routine surveillance basis.

**Key Points**

- Oral cGVHD typically presents first as new-onset dry mouth and oral sensitivity and may resemble mucositis. Mature lesions can develop tissue and salivary gland fibrosis.
- Patients with a history of oral cGVHD have an increased risk of subsequent oral cancer. Thus, early and aggressive treatment is recommended along with regular screening.
- Non-GVHD complications should be vigilantly managed including dental caries, xerostomia, gingivitis, dysgeusia, drug reactions, and oral infections.
- Dentists with expertise in oral medicine should be part of multidisciplinary teams of a long-term HCT follow-up/survivorship clinic.

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